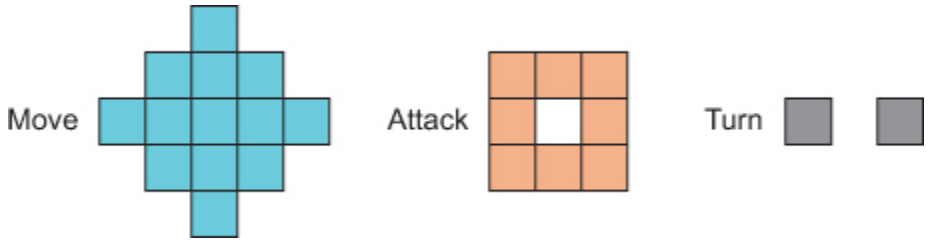
Figure 9.25. This depicts the action space for agents in the MAgent library. Each agent can move in 13 different directions or attack in 8 directions immediately around it. The turn actions are disabled by default, so the action space is 13 + 8 = 21.



Listing 9.15. Finding the neighbors

```
def get_neighbors(j,pos_list,r=6):        1
    neighbors = []
    pos_j = pos_list[j]
    for i,pos in enumerate(pos_list):
        if i == j:
            continue
        dist = cityblock(pos,pos_j)
        if dist < r:
            neighbors.append(i)
    return neighbors
```

- ***1*** Given [x,y] positions of all agents in pos_list, returns indices of agents that are within the radius of agent j

We need this function to find the neighbors in the FOV of each agent to be able to calculate the mean action vector. We can use `env.get_pos(team1)` to get a list of coordinates for each agent on team 1, and then we can pass this into the `get_neighbors` function along with an index, `j`, to find the neighbors of agent `j`.

```
>>> get_neighbors(5,env.get_pos(team1))
[0, 1, 2, 4, 6, 7, 8, 9, 10, 13]
```

So agent `5` has 10 other agents on team 1 within its 13 × 13 FOV.

We now need to create a few other helper functions. The actions that the environment accepts and returns are integers 0 to 20, so we need to be able to convert this to a one-hot action vector and back to integer form. We also